Nuclei Isolation Reagent Kits

Nuclei Isolation Reagents

Nuclei Extraction Buffer	Stored at 4ºC
MACS BSA Stock Medium	Stored at 4ºC
PBS	Stored at 4ºC
RNase Inhibitor	Stored at -20ºC
AOPI dye	Stored at 4 ºC

Nuclei Isolation Consumables

GentleMACs (TM) C-tube	Stored at RT
MACS®SmartStrainers (70µm)	Stored at RT
MACS®SmartStrainers (30µm)	Stored at RT
15mL Eppendorf DNA LoBind tubes	Stored at RT
Sterile Scalpel	Stored at RT
Sterile forceps	Stored at RT

Additional Nuclei Isolation Equipment:

GentleMACS Octo Dissociator
Luna F7X Fluorescent Cell Counter
Microscope (with at least 40x resolution)

Get Started

Prior to starting the experiment, samples should be picked out and analyzed to ensure criteria are met. All centrifugation steps need to be performed at 4°C. All instruments, buffers and consumables with sample contact should be pre-cooled and put on ice. All steps should be done on ice to ensure that integrity of sample is not compromised.

Cutting Snap Frozen Tissue

Pre-cut all tissue samples the day before the experiment or the same day and transfer to a 1.8mL Eppendorf tube. All steps must be completed on dry ice and tissue should never thaw.

1. Obtain tissue sample vial from the -80 freezer or the Liquid Nitrogen Tank and immediately place on dry ice
2. Transfer tumor breast tissue to a pre-chilled petri dish and cut 150-200 mg
   If tissue appears fatty (easy to cut and appears yellow) or crystalized (very hard to cut), additional tissue amount may be needed.
3. Mince frozen tissue into <1mm pieces

Buffer Preparation - Lysis Buffer & Resuspension Buffer:

Prepare the following Lysis and Resuspension Buffers on ice shortly before starting the Nuclei Isolation protocol. Prepare large volumes in a 15-ml or 50-ml conical tube. Vortex briefly before use. *Table below assumes you are starting with a RNAase Inhibitor Stock Concentration of 40 U/µL.*

For lysis buffer RNase Inhibitor (0.2 U/µL final conc.). For Resuspension buffer MACS BSA Stock Solution 1: 250 (4% final conc.), Nuclei Extraction Buffer 1:7 (14% final conc.), RNase Inhibitor (0.2 U/µL final conc.). Table below assumes a RNAase Inhibitor Stock Concentration of 40 U/µL.

Lysis Buffer

Lysis Buffer (4 mL/sample) ***Add reagents in the order listed***	1 sample +10% (µL)	2 samples +10% (µL)	4 samples +10% (µL)	6 samples +10% (µL)	8 samples +10% (µL)
Nuclei Extraction buffer	4,378	8,756	17,512	26,268	35,024
RNAase Inhibitor	22	44	88	132	176
Total	4,400	8,800	17,600	26,400	35,200

Resuspension Buffer

Resuspension Buffer (1.5 mL/sample) ***Add reagents in the order listed***	1 sample +10% (µL)	2 samples +10% (µL)	4 samples +10% (µL)	6 samples +10% (µL)	8 samples +10% (µL)
PBS (1x)	1,399.5	2,799	5,598	8,397	11,196
Nuclei Extraction buffer	235.7	471.4	942.8	1,414.2	1,885.6
BSA Stock Medium	6.60	13.2	26.4	39.6	52.8
RNAase Inhibitor	8.25	16.5	33	49.5	66
Total	1,650	3,300	6,600	9,900	13,200

Nuclei Isolation Protocol

Prior to starting the isolation, set up a station next to the gentleMACS Octo Dissociator by placing the dry ice tray with frozen samples and bucket of wet ice with pre-cooled gentleMACS C tubes

1. Add 2mL of ice-cold lysis buffer to each pre-cooled gentleMACS C Tube and place on ice.
2. Transfer minced tissue pieces to gentleMACS C Tube containing lysis buffer and directly proceed to the following steps until samples are dissociated.
   DO NOT let frozen samples thaw before dissociation as endogenous RNase might degrade RNA.
3. Close gentleMACS C tube and snap it on the gentleMACS dissociator, then place the cooling sleeve over it.
4. Chambers will show as "selected"; press Start to run Program 4C_nuclei_1 on the gentleMACS Octo dissociator.
5. After run is complete incubate the samples on the dissociator for an additional 10 minutes.
6. Detach C tube from the dissociator and place the C tube immediately on ice.
7. Assemble a 70µm MACS®SmartStrainer on a 15mL LoBind Eppendorf tube and using a wide pipette tip, slowly transfer the nuclei suspension from the C-tube onto the strainer.
8. To maximize nuclei recovery, add 2mL of ice-cold lysis buffer to the C tube. Invert the C tube a few times to rinse and transfer contents to the 70µm MACS®SmartStrainer.
9. Discard the 70µm MACS®SmartStrainer.
10. Centrifuge nuclei suspension at 500xg for 10 minutes at 4°C. Carefully discard supernatant using a pipette without disturbing nuclei pellet. Leave behind a small fraction (~200 µl) of supernatant if nuclei pellet is not apparent.
11. Resuspend nuclei pellet with 1.5mL of ice-cold resuspension buffer by gently pipette mix at least 10x, continuing until no pellet can be visualized.
    At this point take out AOPI stain from the fridge to get to room temperature before counting.
12. Assemble a 30µm MACS®SmartStrainer on a new 15mL LoBind Eppendorf tube and using a wide pipette tip, slowly transfer the nuclei suspension onto the strainer.
13. Determine nuclei concentration using AOPI. Using a 0.5mL Eppendorf tube, mix 9µL of the sample with 1µL of AOPI dye, and pipette gently to ensure the suspension is mixed. Carefully add 10µL into one of the grooves of the Photon slide and count using the Luna F7X Automated Cell Counter. **Save the count as A_B_30uM_Nuclei_TUM_Br.**

    A=Sample ID for sample in chamber A, B = Sample ID for sample in chamber B, 30uM=Count After 30 µm Strainer, TUM=tumor, Br= Breast.

14. Determine if nuclei sample requires Debris Removal, and which method is appropriate. If debris is minimal skip this protocol and directly proceed to Scale Fixation Protocol.

Debris Removal Protocol

Prior to starting the debris removal protocol, prepare the appropriate amount of Resuspension Buffer according to the tables below, label accordantly and place on ice.

Resuspension Buffer *with* RNAase Inhibitor

Resuspension Buffer (2.5 mL/sample) *Add reagents in the order listed*	1 sample +10% (µL)	2 samples +10% (µL)	4 samples +10% (µL)	6 samples +10% (µL)	8 samples +10% (µL)
PBS (1x)	2,340.25	4,680.5	9,361	14,041.5	18,722
Nuclei Extraction buffer	385	770	1,540	2,310	3,080
BSA Stock Medium	11	22	44	66	88
RNAase Inhibitor	13.75	27.5	55	82.5	110
Total	2,750	5,500	11,000	16,500	22,000

Resuspension Buffer *without* RNAase Inhibitor

Resuspension Buffer (4 mL/sample) *Add reagents in the order listed*	1 sample +10% (µL)	2 samples +10% (µL)	4 samples +10% (µL)	6 samples +10% (µL)	8 samples +10% (µL)
PBS (1x)	3,766.4	7,532.8	15,065.6	22,598.4	30,131.2
Nuclei Extraction buffer	616	1,232	2,464	3,696	4,928
BSA Stock Medium	17.6	35.2	70.4	105.6	140.8
Total	4,400	8,800	17,600	26,400	35,200

1. Aliquot 1 x 10^6 nuclei into a 5 mL Eppendorf tube by gently pipette mix before aliquoting to ensure accurate number of nuclei.

   Example:
   Concentration (C) = 1.8x10^6 cells/mL
   Target (T) =2x10^5 cells
   Volume = X

$$V = \frac{T}{C} = \frac{0.2x10^6\ cells}{1.8x10^6 cells/mL} = 0.111\ mL = 111\ \mu L$$

2. Centrifuge nuclei suspension at 300xg for 5 minutes at 4°C. Carefully discard supernatant using a pipette without disturbing nuclei pellet. Leave behind a small fraction of supernatant if nuclei pellet is not apparent.
3. Gently resuspend nuclei pellet in 450µL of nuclei resuspension buffer.
4. Add 50µL of Anti-Nucleus Microbeads.
5. Mix well and incubate for 15 minutes in the refrigerator (2-8°C).

6. Add an additional 2mL of resuspension buffer *with* RNase inhibitor to the tube containing the nuclei and proceed to magnetic separation
7. Place pre-chilled column in the magnetic field of a cold suitable MACS separator.
8. Prepare column by rinsing with 3 mL of nuclei suspension buffer *without* RNase inhibitor. Allow buffer to completely soak through the surface of the dark sponge at the center before proceeding.
9. Apply nucleus suspension directly to the center of the column, collecting flow through containing debris.
10. Wash column two times with 1mL of nuclei resuspension buffer without RNase inhibitor and collect debris that passes through in the same tube.
11. Remove column from magnetic separator and place it on a pre-cooled 15mL LoBind Eppendorf Tube.
12. Pipette 1mL of nuclei resuspension buffer *with* RNase inhibitor onto the column. Immediately flush out the magnetically labeled nuclei by firmly pushing the plunger into the column.
13. Determine final nuclei concentration using AOPI. Using a 0.5mL Eppendorf tube, mix 9µL of the sample with 1µL of AOPI dye, and pipette gently to ensure the suspension is mixed. Carefully add 10µL into one of the grooves of the Photon slide and count using the Luna F7X Automated Cell Counter. **Save the count as A_B_ADR_Nuclei_TUM_Br.**

    A=Sample ID for sample in chamber A, B = Sample ID for sample in chamber B, ADR=After Debris Removal Count, TUM=tumor, Br=Breast.

14. Calculate amount of nucleus suspension needed to start with 200,000 cells per sample for fixation.

    Example:
    Concentration (C) = $4x10^5$ cells/mL
    Target (T) =$2x10^5$ cells
    Volume = X

$$V = \frac{T}{C} = \frac{2x10^5\ cells}{4x10^5 cells/mL} = 0.5\ mL = 500\ \mu L$$

ScalePlex Fixation Protocol:

This protocol shows the steps for fixation of 8-12 samples. <u>50,000 to 200,000 nuclei per sample are required</u> to proceed with this protocol. DO NOT proceed if counts are lower

ScalePlex Reagent Kits

ScalePlex Fixation Reagents

Wash Buffer	Stored at -20ºC
Fixation Reagent	Stored at 4ºC
ScalePlex Oligo Plate	Stored at -20ºC
1x PBS	Stored at 4 ºC
DEPC	Stored at 4 ºC
DMSO	Stored at RT
Methanol	Stored at RT
AOPI	Stored at 4 ºC

ScalePlex Consumables

15-mL conical tubes	Stored at RT
5-mL DNA LoBind tubes	Stored at RT
1.5-mL DNA LoBind tubes	Stored at RT
0.5-mL Protein LoBind tubes	Stored at RT
96-well LoBind semi-skirted PCR plates	Stored at RT
Microseal 'B' PCR Plate Sealing Film	Stored at RT

Get Started

Prior to starting the fixation protocol take out the fixation reagent from the 4ºC and place at room temperature. Remove Wash Buffer from -20ºC, once thawed, invert Wash Buffer to ensure it is fully mixed, then place on ice. Remove ScalePlex Oligo Plate from -20ºC and place on ice. Aliquot 2 mL of ice-cold methanol to a 5mL Eppendorf Tube.

Nuclei Preparation & ScalePlex Fixation

1. Centrifuge samples at 500 xg for 10 minutes at 4°C
2. While centrifuging, calculate the fixation target concentration volume required for ScalePlex for the entire sample using the table below:

Target Cell Input per Sample for Fixation	Target Concentration	Volume required for Fixation
50,000	2,000 cells/µL	25 µL
100,000	4,000 cells/µL	25 µL
200,000	8,000 cells/µL	25 µL

Example: (*If desired concentration is 8000 cell/uL for. Divide the total concentration of your sample by 8000)*

Current Concentration (C_c) = 4x10^5 cells/mL
*Target Concertation (C_t) = 8,000 cells/*µL
Volume = X

$$V = \frac{C_c}{C_t} = \frac{4x10^5 cells/mL}{8000\ cells} = 0.05\ mL = 50\ \mu\text{L}$$

3. At this point you can start preparing the fixation reagent. Start by briefly spinning down the room temperature fixation reagent. In a chemical fume hood, add 50 µL DMSO to the Fixation Reagent Tube.
4. Vortex at high speed with intermittent brief spins to dissolve the Fixation Reagent. This may take up to several minutes. Ensure the pellet is fully dissolved before proceeding.
5. Briefly spin down the Fixation Reagent tube.
6. On ice, prepare Fixation Solution by adding the fixation reagent with DMSO to the 5mL DNA LoBind tube containing the 2mL of ice-cold methanol.
7. Briefly vortex the Fixation Solution mix and leave on ice. Use within 6 hours.
8. Once centrifugation is complete, carefully remove supernatant by pipetting without disturbing the pellet
9. Dilute nuclei with ice cold 1X PBS to the required target concentration calculated in step 2
10. Resuspend the nuclei by flicking. Proceed immediately to ScalePlex Fixation (for 8-12 samples).

11. Transfer 25uL of diluted cells to an Eppendorf twin.tec PCR LoBind plates 96 well plate on ice. Record the wells used in the PCR plate and their corresponding sample ID.
12. Spin down the ScalePlex Oligo Plate by pulsing until it reaches 1000 xg at 4°C and place on ice.
13. Using a sharp tool, remove the seal from the appropriate number of ScalePlex Oligo wells to be used.
14. Transfer 5 µL of oligos from the ScalePlex Oligo Plate to each well of the PCR plate containing the nuclei samples. Record the ScalePlex Oligo used for each sample.
15. Set pipette to 25uL and gently pipette up and down five times to mix using a multi-channel or single channel pipettor
16. Incubate samples on ice for 5 minutes.
17. Store ScalePlex Oligo Plate containing unused oligos at -20°C.
18. Immediately before use, in the *chemical fume hood*, add 20 uL of DEPC to the Complete Fixation Reagent.
19. Briefly vortex the Complete Fixation Solution to mix.
20. In the hood, add 120 µL of the Complete Fixation Solution slowly to each tube containing the ScalePlex stained cells. Gently pipette up and down five times to mix.
21. Incubate on ice for 15 minutes.

Number of Samples	8	9-12
Container Size	5ml LoBind tube	15 ml LoBind tube
Volume of Wash Buffer for first wash	2.4 mL	4 mL

22. While incubating add the following volume of Wash Buffer to the appropriate tube and keep in ice.

23. After incubation, use a single-channel pipette to pool samples by transferring the cell suspension to the appropriate tube containing Wash Buffer. Same tip can be used.
24. Centrifuge the tube containing samples in Wash Buffer at 500 x g for 10 minutes at 4°C.
25. If samples appeared to be debris heavy, proceed to step 26. If samples did not appear debris heavy, proceed to step 31.
26. Carefully remove supernatant without disturbing the pellet, leaving ~200 µL of residual volume.
27. Resuspend the nuclei by flicking the tube and then add 800 µL ice cold Wash Buffer to the cells.
28. Add an additional 4 mL of ice-cold Wash Buffer to the nuclei.
29. Centrifuge the tube at 500 x g for 10 minutes at 4ºC.

30. If pellet is visible, proceed with the next wash. If pellet is barely visible, proceed to step 34.
31. Carefully remove supernatant without disturbing the pellet, leaving ~50 µL of residual volume.
32. Resuspend the nuclei by flicking the tube and then add 1 mL ice cold Wash Buffer to the cells.
33. Centrifuge the tube at 500 x g for 10 minutes at 4°C.
34. Carefully remove supernatant without disturbing the pellet, leaving ~50 µL of residual volume.

Number of Samples	8	9-12
Volume of Wash Buffer for resuspension	30 µL	50 µL
Final Volume	80 µL	100 µL

35. Resuspend the nuclei by flicking the tube and add the appropriate volume of ice-cold Wash Buffer gently.
36. Determine the concentration of the cell suspension using cell counting equipment. Using a 0.5 mL lobind tube, add 18 µL of ice-cold PBS and 2 µL of the nuclei suspension. Pipette mix and transfer 9 µL of the diluted nuclei to a new 0.5 mL tube. Add 1uL of AOPI dye. Mix and load 10 µL in the Luna slide.
37. Record nuclei concentration and calculate desired suspensions to freeze which require a concentration between 3,000-10,000 cells/µL.

Example: *(If desired concentration is 10,000 cell/uL)*

Current Concentration (C_c) = 1.2x1^6 cells/mL
Target Concertation (C_t)= 10,000 cells/uL
Dilution factor (DF) = 1:10
Current volume (V_c) = 80uL = 0.08mL
Total Volume Needed = X

$$V = \frac{C_c \times DF \ \times V_c}{C_t} = \frac{1.2x10^6 cells/mL \times 10 \times 0.08mL}{10{,}000\ cells} = \mathbf{96\ \mu L}$$

Now Subtract the Total Volume needed minus the current volume, to obtain the volume needed to achieve the desired concentration.

$$V = 96 - 80 = \mathbf{16\ \mu L}$$

38. Store the fixed nuclei in a pre-labeled Scale 1.5 Eppendorf, place tube in the Scale Box in the -80°C freezer and record the scale poll ID.